

cloudChronicles Lab #001: Disaster Recovery Detective

Lab Type: Idea

Estimated Time: 30–45 mins

Skill Level: Beginner

```
# Let's begin by printing your name to personalize the notebook
your_name = "Mr. Raheem Kareem"
print(f"Welcome to the lab, {your_name}!")
```

Welcome to the lab, Mr. Raheem Kareem!

STAR Method Lab Prompt

Situation:

Imagine our company runs a high-traffic app like Instagram or TikTok hosted entirely in Google Cloud's us-south1 region (Dallas, Texas). Suddenly, a severe weather event or massive power grid failure causes a regional outage—meaning the entire us-south1 data center goes completely offline.

Without a recovery plan:

Users face broken screens and "500 Internal Server Errors."

Data saved only in Texas becomes temporarily inaccessible.

The company loses user trust and thousands of dollars every minute it remains offline.

Task:

As the Lead Cloud Architect, my task is to design a high-availability Disaster Recovery (DR) Plan. This plan must automatically detect when us-south1 goes down, route all traffic over to a safe secondary location (us-east1), protect all user data from loss, and automatically transition traffic back once Texas comes back online—all while keeping downtime as close to zero as possible.

Action:

Part 1: Setup & Proactive Redundancy Before any disaster hits, we build our safety net using three core Google Cloud tools: Database Safety (Cloud SQL Cross-Region Replicas): Analogy: Imagine writing in a digital notebook that automatically copy-pastes every letter into a twin notebook kept in another state in real time. How it works: Our main primary database lives in us-south1. We set up a Cross-Region Read Replica in us-east1. As soon as data is saved in Texas, Google Cloud syncs it over to South Carolina instantly. File Storage Safety (Multi-Region Cloud Storage): Analogy: Storing your photos in a universal locker instead of on a single hard drive. How it works: We configure our user upload buckets to use Multi-Region Cloud Storage (US). Instead of locking files inside Texas, Google Cloud automatically mirrors every picture and video across multiple geographic locations across the US. Monitoring & Alarm System (Pub/Sub + Cloud Monitoring): Analogy: A smoke detector wired directly to an automated sprinkler system. How it works: Cloud Monitoring continuously checks our servers' pulse every few seconds. If it detects a crash, it instantly publishes a message to a Pub/Sub topic (a cloud messaging system), which fires off alerts to our engineers' phones and triggers automated repair scripts.

Part 2: Failover & Recovery Process Phase A: The Failover (When Disaster Strikes) 1.1. Outage Detection: Automated Monitoring.us-south1 goes dark. Cloud Monitoring fails 3 consecutive health checks and immediately sends an emergency payload to the Pub/Sub topic. 2.2. Automated Alert & Script Trigger: Zero Manual Delay. The Pub/Sub alert fires a Cloud Function (an automated script), notifying the engineering team on Slack and executing the failover sequence. 3.3. Promote the Replica: Cloud SQL Promotion. The Cloud Function promotes the us-east1 Cloud SQL Read Replica into the new Primary Read/Write database. 4.4. Reroute Web Traffic: Global Cloud Load Balancer. Google Cloud's Global Load Balancer updates its routing rules. It stops sending web traffic to Texas and immediately redirects all user requests to us-east1.

Phase B: The Recovery (When Texas Comes Back Online) Re-syncing the Primary Region: Once Google restores power to us-south1, we do not jump back immediately. We first set up a new read replica in us-south1 to catch up on all new data created in us-east1 during the outage.

Planned Failback: During off-peak hours (e.g., 2:00 AM), we route traffic back to us-south1 smoothly using the Global Load Balancer.

Restoring Baseline: us-south1 resumes its job as the main home base, and us-east1 returns to being the standby backup location.

Expected Result:

By implementing this disaster recovery plan, our system achieves:

Near-Zero Data Loss (RPO ~ 0): Thanks to real-time Cloud SQL replication and Multi-Region Cloud Storage, user data remains intact.



Rapid Recovery (RTO < 5 minutes): Automated Pub/Sub triggers and Load Balancers flip the switch long before engineers could manually log in to fix it.

Business Continuity: Users around the world continue using the application without ever knowing a major regional data center went offline.

T **B** **I** **<>** **↔** **🖼️** **”** **☰** **☰** **—** **ψ** **😊** **☰** [Close](#)

📌 Your Assignment

Use this section to complete your deliverable:

```markdown

(Example Format)

- **Primary Region**: us-south1 (Dallas, TX)
- **Backup Location**: us-east1 (South Carolina)
- **Failover Trigger**: Health checks fail 3 consecutive times fires an automated event
- **Redundancy Services**:
  - Cloud SQL Cross-Region Replicas: Read-only clone running on us-east1.
  - Multi-Region Cloud Storage: Automatically duplicates files across regions simultaneously.
  - Pub/Sub + Cloud Functions: Automated warning system and failover orchestrator.
- **Backup Schedule**: Continuous real-time database replication multi-region snapshot backups stored for 30 days.

```



Your Assignment

Use this section to complete your deliverable:

(Example Format)

- **Primary Region**: us-south1 (Dallas, TX)
- **Backup Location**: us-east1 (South Carolina)
- **Failover Trigger**: Health checks fail 3 consecutive times
- **Redundancy Services**:
 - Cloud SQL Cross-Region Replicas: Read-only clone running
 - Multi-Region Cloud Storage: Automatically duplicates files
 - Pub/Sub + Cloud Functions: Automated warning system and failover
- **Backup Schedule**: Continuous real-time database replication